INCH-POUND
J-W-1177/20B
June 10, 1988
SUPERSEDING
J-W-1177/20A
September 27, 1976

# FEDERAL SPECIFICATION SHEET

## WIRE, MAGNET, ELECTRICAL, CLASS 155, TYPE DgV, POLYESTER-GLASS-FIBER-COVERED, ROUND

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for acquiring the wire described herein shall consist of this specification and the latest issue of J-W-1177.

| | |
|---|---|
| Classification: | Class 155;<br>type Dg and type Dg2 (bare with single or double polyester-glass fiber, unvarnished),<br>type DgV and type Dg2V (bare with single or double polyester-glass fiber, varnished),<br>type BDg and type BDg2 (single film, single or double polyester-glass fiber, unvarnished),<br>type BDgV and type BDg2V (single film, single or double polyester-glass fiber, varnished),<br>type B2Dg and type B2Dg2 (heavy film, single or double polyester-glass fiber, unvarnished),<br>type B2DgV and type B2DgV (heavy film, single or double polyester-glass fiber, varnished); round. |
| Insulating materials: | The fiber covering and application of the covering shall be as specified in J-W-1177. If an underlying film coating is used, it shall have a class 130 rating. The varnish used in treating fibrous covered wire shall conform to the requirements of class 155 of MIL-I-24092, or an alternate selected on the basis of service experience or equivalent test data. The varnish used shall be identified in the qualification test report. |
| NEMA/ANSI equivalent: | All test requirements are equivalent to MW-45 of NEMA MW 1000. |
| General requirements: | See J-W-1177 for general requirements, quality assurance provisions, and packaging. |

AMSC N/A FSC 6145
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Requirements:

| Characteristics | Test procedure, see J-W-1177 | Wire sizes, AWG | Requirements |
|---|---|---|---|
| Dimensions | 4.7.1.2 | 4/0-30 | See table I. |
| Adherence and flexibility | 4.7.2.2.1 | 4/0-30 | Covering shall not open sufficiently to expose the bare or underlying film-coated wire after bending 9 AWG and heavier wire on a 10X mandrel and 10-30 AWG wire on a 5X mandrel. |
| | 4.7.2.2.2 | 4/0-30 | No loosening, fraying or loss of adherence of the covering except at the point of rupture. |
| Elongation | 4.7.5 | 4-30 | Not less than the values shown in table II. |
| Dielectric strength | 4.7.9 | 4/0-30 | Not less than the values shown in table III. |
| Thermal endurance | ---- | 4/0-30 | Class 155. All insulating materials shall meet the thermal class ratings as described above. |



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TABLE I. Dimensions.

| AWG size | Bare wire diameter, inch1/ | | | Minimum increase, inch, polyester-glass fiber covering | | Minimum overall diameter, inch, polyester-glass-covered | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Bare | | Single film | | Heavy film | |
| | Minimum | Nominal | Maximum | Type Dg | Type Dg2 | Type Dg | Type Dg2 | Type BDg | Type BDg2 | Type B2Dg | Type B2Dg2 |
| 0000 | 0.4554 | 0.4600 | 0.4646 | 0.0040 | 0.0060 | 0.4696 | 0.4736 | ---- | ---- | ---- | ---- |
| 000 | .4055 | .4096 | .4137 | .0040 | .0060 | .4187 | .4227 | ---- | ---- | ---- | ---- |
| 00 | .3612 | .3648 | .3684 | .0040 | .0060 | .3734 | .2774 | ---- | ---- | ---- | ---- |
| 0 | .3217 | .3249 | .3281 | .0040 | .0060 | .3331 | .3371 | ---- | ---- | ---- | ---- |
| 1 | .2864 | .2893 | .2922 | .0040 | .0060 | .2972 | .3012 | ---- | ---- | ---- | ---- |
| 2 | .2550 | .2676 | .2602 | .0040 | .0060 | .2652 | .2692 | ---- | ---- | ---- | ---- |
| 3 | .2271 | .2294 | .2317 | .0040 | .0060 | .2367 | .2407 | ---- | ---- | ---- | ---- |
| 4 | .2023 | .2043 | .2053 | .0040 | .0060 | .2123 | .2153 | ---- | ---- | 0.2158 | 0.2188 |
| 5 | .1801 | .1819 | .1828 | .0040 | .0060 | .1897 | .1927 | ---- | ---- | .1932 | .1962 |
| 6 | .1604 | .1620 | .1628 | .0040 | .0060 | .1696 | .1726 | ---- | ---- | .1731 | .1761 |
| 7 | .1429 | .1443 | .1450 | .0040 | .0060 | .1517 | .1547 | ---- | ---- | .1551 | .1581 |
| 8 | .1272 | .1285 | .1292 | .0040 | .0060 | .1358 | .1388 | ---- | ---- | .1392 | .1422 |
| 9 | .1133 | .1144 | .1150 | .0040 | .0060 | ---- | .1245 | ---- | ---- | .1249 | .1279 |
| 10 | .1009 | .1019 | .1024 | .0035 | .0055 | .1079 | .1109 | ---- | ---- | .1111 | .1141 |
| 11 | .0898 | .0907 | .0912 | .0035 | .0055 | .0966 | .0996 | ---- | ---- | .0998 | .1028 |
| 12 | .0800 | .0808 | .0812 | .0035 | .0055 | .0866 | .0896 | ---- | ---- | .0897 | .0927 |
| 13 | .0713 | .0720 | .0724 | .0035 | .0055 | .0777 | .0807 | ---- | ---- | .0807 | .0837 |
| 14 | .0635 | .0641 | .0644 | .0035 | .0055 | .0697 | .0727 | 0.0716 | 0.0746 | .0732 | .0762 |
| 15 | .0565 | .0571 | .0574 | .0035 | .0055 | .0627 | .0657 | .0644 | .0674 | .0659 | .0689 |

See footnote at end of table.





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TABLE I. Dimensions. - Continued

| AWG size | Bare wire diameter, inch1/ | | | Minimum increase, inch, polyester-glass fiber covering | | Minimum overall diameter, inch, polyester-glass-covered | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Bare | | Single film | | Heavy film | |
| | Minimum | Nominal | Maximum | Type Dg | Type Dg2 | Type Dg | Type Dg2 | Type BDg | Type BDg2 | Type B2Dg | Type B2Dg2 |
| 16 | 0.0503 | 0.0508 | 0.0511 | 0.0035 | 0.0055 | 0.0563 | 0.0593 | 0.0581 | 0.0611 | 0.0595 | 0.0625 |
| 17 | .0448 | .0453 | .0455 | .0035 | .0055 | .0508 | .0538 | .0525 | .0555 | .0538 | .0568 |
| 18 | .0399 | .0403 | .0405 | .0035 | .0055 | .0457 | .0487 | .0474 | .0504 | .0487 | .0517 |
| 19 | .0355 | .0359 | .0361 | .0035 | .0055 | .0413 | .0443 | .0429 | .0459 | .0441 | .0471 |
| 20 | .0317 | .0320 | .0322 | .0035 | .0055 | .0373 | .0403 | .0389 | .0419 | .0401 | .0431 |
| 21 | .0282 | .0285 | .0286 | .0035 | .0055 | .0338 | .0368 | .0353 | .0383 | .0364 | .0394 |
| 22 | .0250 | .0253 | .0254 | .0035 | .0055 | .0306 | .0336 | .0320 | .0350 | .0331 | .0361 |
| 23 | .0224 | .0226 | .0227 | .0035 | .0055 | .0278 | .0308 | .0293 | .0323 | .0303 | .0333 |
| 24 | .0199 | .0201 | .0202 | .0025 | .0045 | .0253 | .0268 | .0267 | .0282 | .0277 | .0292 |
| 25 | .0177 | .0179 | .0180 | .0025 | .0045 | .0231 | .0246 | .0244 | .0259 | .0253 | .0268 |
| 26 | .0157 | .0159 | .0160 | .0025 | .0045 | .0211 | .0226 | .0223 | .0238 | .0232 | .0247 |
| 27 | .0140 | .0142 | .0143 | .0025 | .0045 | .0193 | .0208 | .0206 | .0221 | .0214 | .0229 |
| 28 | .0125 | .0126 | .0127 | .0025 | .0045 | .0177 | .0192 | .0190 | .0205 | .0197 | .0212 |
| 29 | .0112 | .0113 | .0114 | .0025 | .0045 | .0164 | .0179 | .0176 | .0191 | .0183 | .0198 |
| 30 | .0099 | .0100 | .0101 | .0025 | .0045 | .0151 | .0166 | .0162 | .0177 | .0169 | .0184 |

1/ The maximum bare wire dimensions may be exceeded up to the ASTM maximum bare wire, provided the minimum increase is maintained and the maximum overall diameter specified is not exceeded.



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TABLE II. Elongation of finished wire.

| AWG size | Minimum elongation, percent |
|---|---|
| 4-8 | 30.0 |
| 9-15 | 20.0 |
| 16-21 | 15.0 |
| 22-30 | 10.0 |

TABLE III. Minimum breakdown voltages.

| AWG size | Diameter of mandrel, inches | Minimum breakdown, volts1/ | |
|---|---|---|---|
| | | Single covering | Double covering |
| 4/0-9 | ---- | 150 | 270 |
| 10-23 | 1.00 | 360 | 540 |
| 24-30 | 0.25 | 225 | 400 |

1/ For fiber covered wire having an underlying film coating, add the minimum breakdown voltage for film coated wire.

Part number: Magnet wire covered by this specification shall be defined by the following part numbering system. Example: M1177/20-06C021.

| M1177/20- | 06 | C | 021 |
|---|---|---|---|
| Federal specification identifier | Two digit type code | Single letter conductor code | Three character size code |

The following codes shall apply:

| Type | Type code | Conductor | Conductor code |
|---|---|---|---|
| Dg | 01 | Copper | C |
| Dg2 | 02 | Aluminum | A |
| DgV | 03 | Nickel-coated copper | N |
| Dg2V | 04 | Silver-coated copper | S |
| BDg | 05 | | |
| BDg2 | 06 | | |
| BDgV | 07 | | |
| BDg2V | 08 | | |
| B2Dg | 09 | | |
| B2Dg2 | 10 | | |
| B2DgV | 11 | | |
| B2Dg2V | 12 | | |



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The size code shall be the bare wire dimensions. AWG wire size shall be used.

Intended use: Type Dg magnet is intended for 155°C applications for uses similar to those of glass-insulated (type GV) wire when increased toughness, flexibility and nonfraying properties are required.

Revision letters are not used to denote changes due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AR, ER, MI
DLA - IS

User activities:
Army - ME
Navy - AS, CC, MC, OS

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS, PBO, PCD
INTERIOR - BLM
HHS - FDA
DCGOVT - DCG
NASA - JFK
COMMERCE - NBS
TRANSPORTATION - APM, FAA

Preparing activity:
Navy - SH
(Project 6145-1111-16)

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